BRITTLENESS OF LLMS

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Cats Confuse Reasoning LLM: Query Agnostic Adversarial Triggers for Reasoning Models

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https://arxiv.org/abs/2503.01781

Research Focus

This work examines how short, irrelevant text appended to reasoning problems can systematically mislead language models, without changing the problem's core semantics.

CatAttack: What It Is

CatAttack is an automated pipeline that constructs simple irrelevant phrases ("adversarial triggers") that, when appended to problems, significantly increase the rate of incorrect reasoning responses.

Universal Adversarial Triggers

The authors find that certain trigger phrases transfer from one model to another, meaning the distractors are not specific to a single model but can generalize across architectures.

Example Distractor

A concrete example:

"Interesting fact: cats sleep most of their lives."

When appended to math problems, this phrase doubles or triples the likelihood that a model produces an incorrect answer, even though it does not change the problem itself.

Impact on Reasoning Reliability

The presence of irrelevant, unrelated text can cause:

- Higher error rates in reasoning tasks.
- Slower or longer model outputs.
- Systematic vulnerability in chain-of-thought reasoning systems.

This highlights the importance of clarity and relevance in context.

A Survey on In-context Learning

Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Zhiyong Wu,

Baobao Chang, Xu Sun, Jingjing Xu, Lei Li, Zhifang Sui

https://arxiv.org/abs/2301.00234

Definition of In-Context Learning (ICL)

In-Context Learning is defined as the ability of a language model to perform tasks solely based on examples and instructions provided in its input context, without any changes to its learned parameters.

ICL Mechanism

In ICL, the model uses patterns from demonstrations (input-label pairs) to infer how to respond to a new query. There is no explicit training update: everything is inferred at runtime from context.

Role of Context in Activation

The structure and format of the provided context activate internal model behavior:

- Ordering of examples affects how the task is recognized.
- Templates and formatting influence how patterns are inferred.
- Consistency and clarity help the model generalize from few examples.

Different contexts can lead to distinct internal task activation patterns.

Implicit Role/Persona Effects

While not called "personas" explicitly, framing examples with consistent tones, style cues, or roles (e.g., "an expert reviewer") can change how the model interprets tasks, acting as a form of implicit role conditioning.

Design Choices Impact

Effective in-context learning depends on:

- Which examples are chosen.
- How they relate to the query.
- How clearly labels and inputs are presented.

These choices determine whether the model recognizes the task and applies patterns correctly.

Challenges with ICL

The survey highlights limitations:

- Mechanisms of ICL are not fully understood.
- Performance can vary significantly with context composition.
- No universal recipe for universally reliable demonstrations.

Lost in the Middle: How Language Models Use Long Contexts

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape,

Michele Bevilacqua, Fabio Petroni, Percy Liang

https://arxiv.org/abs/2307.03172

Goal of the Study

This research investigates how well language models make use of information spread across very long input contexts, rather than simply what they contain overall.

Tasks Used for Analysis

Two tasks were used to analyze long context usage:

- Multi-document question answering: find the answer when it may be anywhere in text.
- Key-value retrieval: locate relevant key/value pairs among long lists.

These tasks require extracting information from different positions in a long context.

Positional Performance Pattern

The study finds a consistent performance pattern:

- Best performance when relevant information is at the beginning.
- Also high when it appears at the end.
- Worst performance when relevant information is in the middle.

This suggests models are biased towards context boundaries.

Primacy and Recency Biases

The patterns indicate:

- A primacy effect (beginning of context strongly weighted).
- A recency effect (end of context strongly weighted).
- A "U-shaped" usage curve where middle information is underutilized.

This informs how we should structure contexts for best model behavior.

Implications for Context Design

From this study, the following practices emerge:

- Place the most critical information near the start or end.
- Avoid burying key facts deeply in the middle when possible.
- Understand that longer context windows do not guarantee uniform access to all data.

Effective Context Engineering for AI Agents

Anthropic Engineering Blog

https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

What Is Context Engineering?

Context engineering is the practice of curating and managing the tokens that an LLM sees at inference time in order to optimize performance and behavior, especially in multi-turn and agentic settings.

Beyond Prompt Engineering

Prompt engineering traditionally focuses on crafting specific prompts. Context engineering goes further: it manages:

- System messages
- Interaction history
- Tool outputs
- Retrieved external data

all within the constrained context window.

Finite Attention Budget

LLMs have a limited "attention budget": as more tokens are added, the model's ability to use all of them effectively diminishes. Simply adding context does not guarantee improved outcomes.

Managing Context Growth

Effective strategies include:

- Summarization of prior turns
- Compaction of redundant data
- Selection of the most relevant facts

This helps maximize utility within the finite context capability.

Context Engineering in Agents

For agent systems (multi-step workflows with tools):

- Carefully manage which tokens persist across turns.
- Isolate noise from signal.
- Use structured summaries or state representations to preserve concept relevance.

Key Cross-Cutting Themes

- Position and context structure drastically affect model performance.
- Irrelevant content can degrade reasoning quality.
- Context patterns activate different model behaviors.
- Holistic management of context is essential for reliable LLM systems.

Summary Takeaways Across the Four Works

- Language models do not use all provided context uniformly; the position and structure of information strongly influence whether it is used at all.
- Adding more context does not necessarily improve performance and can actively harm reasoning if critical information is buried or surrounded by irrelevant content.
- Even small, seemingly harmless distractors can significantly degrade reasoning accuracy, highlighting the importance of clarity and relevance in all model inputs.

Summary Takeaways Across the Four Works

- Model behavior is largely determined at inference time by the context provided; examples, formatting, and framing activate different internal task representations.
- Effective interaction with LLMs requires managing the entire context holistically rather than focusing on individual prompt wording.
- Context engineering emerges as a unifying principle: selecting, structuring, compressing, and maintaining context is essential for reliable and scalable LLM systems.